Sofia Salazar-Magaña
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Education

PhD Student in Quantitative and Computational Biosciences 2026 –
Baylor College of Medicine, Houston, TX, USA.

Bachelor's degree in Genomic Sciences 2020 – 2024
National Autonomous University of Mexico (UNAM), Querétaro, México.

Research Experience

Data analyst October 2024 – April 2026
Department of Medicine, Section of Genetic Medicine, The University of Chicago, Chicago, IL, USA.
PI: Hae Kyung Im, PhD

- Applied a novel lab-developed method TFXcan, to leverage deep learning-based DNA-to-epigenome predictions to investigate TF-binding alterations associated with breast cancer risk.
- Applied the scPrediXcan framework in two projects: (1) training deep-learning models for alternative polyadenylation (APA) prediction, and (2) modeling gene expression across dGTEx developmental contexts.

Undergraduate research assistant July 2023 – October 2024
The University of Chicago, Remotely.
PI: Hae Kyung Im, PhD

- Compared the performance of Borzoi and Enformer deep learning models for DNA sequence-based prediction of epigenomic features.
- Provided technical support for lab-developed software enabling TWAS analyses.

Undergraduate research assistant January 2022 – October 2024
International Laboratory for Human Genome Research, National Autonomous University of Mexico, Querétaro, México.
PI: Alejandra Medina-Rivera, PhD

- Analyzed RNA-seq data from the Mexican Lupus Registry (LupusMX) to identify key differentially expressed genes and assembled gene regulatory networks, contributing novel insights into the disease etiology within an understudied population.

Contributions

Adeluwa, T., **Salazar-Magaña, S.**, Sumner S., Gona, S., Nyasimi, F., Carbonetto, P., Baca, S. C., Freedman, M., Gusev, A., Liu, B., Madduri, R., Stephens, M., Gao, G., Amariuta, T., & Im, H. K. (2026). TFXcan reveals transcription factor programs driving GWAS loci. *Cell Genomics*, 101398. https://doi.org/10.1016/j.xgen.2026.101398

Foguet, C., Gil, L., Xu, Y., **Salazar-Magaña, S.**, Ritchie, S. C., Persyn, E., Im, H. K., Inouye, M., & Lambert, S. A. (2026). OmicsPred as a centralized resource for genetic prediction of multi-omic traits. *Nature Genetics*. https://doi.org/10.1038/s41588-026-02726-4

Zhou, Y., Adeluwa, T., Zhu, L., **Salazar-Magaña, S.**, Sumner, S., Kim, H., Gona, S., Nyasimi, F., Kulkarni, R., Powell, J. E., Madduri, R., Liu, B., Chen, M., & Im, H. K. (2025). scPrediXcan integrates deep learning methods and single-cell data into a cell-type-specific transcriptome-wide association study framework. *Cell Genomics*, *5*(5), 100875. https://doi.org/10.1016/j.xgen.2025.100875

Hernández-Ledesma A. L., Coss-Navarrete, E. L., **Salazar-Magaña, S.**, Ramírez-Espinosa, D., Tinajero-Nieto, L., Torres-Valdez, E., Peña-Ayala, A. H., Félix-Rodriguez, G., Frontana-Vázquez, G., García Sotelo, J. S., Thomas-Chollier, M., Trynka, G., Rosetti, F, Fernandez-Valverde, S. L., Gutiérrez-Arcelus, M., Alpízar-Rodríguez, D. & Medina-Rivera, A. (2026). Gene expression profiling of dendritic cell tolerance dysfunction in women with systemic lupus erythematosus. *Frontiers in Immunology 17*, 1771959. https://doi.org/10.3389/fimmu.2026.1771959

Zhou, Y., Sumner, S., Adeluwa, T., Zhu, L., **Salazar-Magaña, S.**, Kim, H., Gona, S., Nyasimi, F., Kulkarni, R., Powell, J., Madduri, R., Liu, B., Chen, M., & Im, H. K. (2026). Protocol to perform cell-type-specific transcriptome-wide association study using scPrediXcan framework. *STAR Protocols, 7*(1), 104306. https://doi.org/10.1016/j.xpro.2025.104306

Ogonowski, N. S., Cao, F., Flores-Ocampo, V., **Salazar-Magaña, S**., Seviiri, M., Song, L., Kugelman, J., MacGregor, S., Cuellar-Partida, G., Gharahkhani, P., Im, H. K., Harding, I. H., Yang, J., Nayler, S., Kumar, K. R., Martin, N. G., Diaz-Torres, S., & Rentería, M. E. GWAS Meta-Analysis of Essential Tremor Uncovers 47 Novel Genetic Loci and Biological Pathways Linked to Brain Morphometry of the Ventral Diencephalon and Cerebellum. **Preprint**. https://doi.org/10.1101/2025.11.10.25339873

Conferences

The Developmental GTEx Resource Enables the Discovery of Disease-Associated Genes. (Talk). *American Society of Human Genetics (ASHG) Annual Conference,* 2026.

The Developmental GTEx Resource Enables the Discovery of Disease-Associated Genes. (Talk). *The 2026 Annual Meeting of the International Genetic Epidemiology Society (IGES).*

Linking transcription factor binding variation to overall and subtype-specific breast cancer risk variants using a deep learning-based approach. (Poster). *American Society of Human Genetics (ASHG) Annual Conference*, 2025.

Deep learning predicts TF binding variation, illuminating breast and ovarian cancer biology. (Poster). *American Society of Human Genetics (ASHG) Annual Conference,* 2024.

Lupus peripheral blood meta-analysis with Recount3 & Recount3-like processed RNA-seq data. (Poster). *ISCB-LA SoIBio BioNetMX International Conference on Bioinformatics,* 2022.

Teaching Experience

TA at Deep Learning in Genomics course, *The University of Chicago.* Spring ‘26 & ‘25
TA at Quantitative Genomics Training Boot Camp, *Columbia University.* Summer ‘25 & ‘24
Lecturer at bioinformatics institutional courses, *International Laboratory for Human Genome Research.* 2024 & 2023

Certifications & Relevant courses

- **Intro to AI-driven Science on Supercomputers,** *Argonne National Laboratory.*
  Credential ID: a1ff9ac49d6d6754ce667fae36f8ad83f8fe93a5
- **Google’s Data Analytics Professional Certificate**
  Credential ID: KEBDX9FHLDSG
- **Workshop: Analysis of biological and biomedical transcriptomic data using Artificial Intelligence (AI) and Machine Learning (ML) methods with R,** *ISCB-LA SoIBio BioNetMX International Conference on Bioinformatics, 2022.*
- **Workshop: Introduction to the logical modeling of cellular networks, with applications to cancer and immunology**, *International Laboratory for Human Genome Research.*

Skills

- Programming (R, Python, Bash, C++, Git, PHP, SQL, Perl).
- Bioinformatics (bulk and single-cell RNA-seq, genome alignment, epigenomic data analysis e.g ChIP-seq, ATAC-seq).
- Deep learning and acceleration hardware: Pytorch, Tensorflow, CUDA, Intel XPU, MPS.
- High Performance Computing (SGE, SLURM).

Languages

Spanish (Native) | English (Proficient – TOEFL iBT 118) | French (Basic)